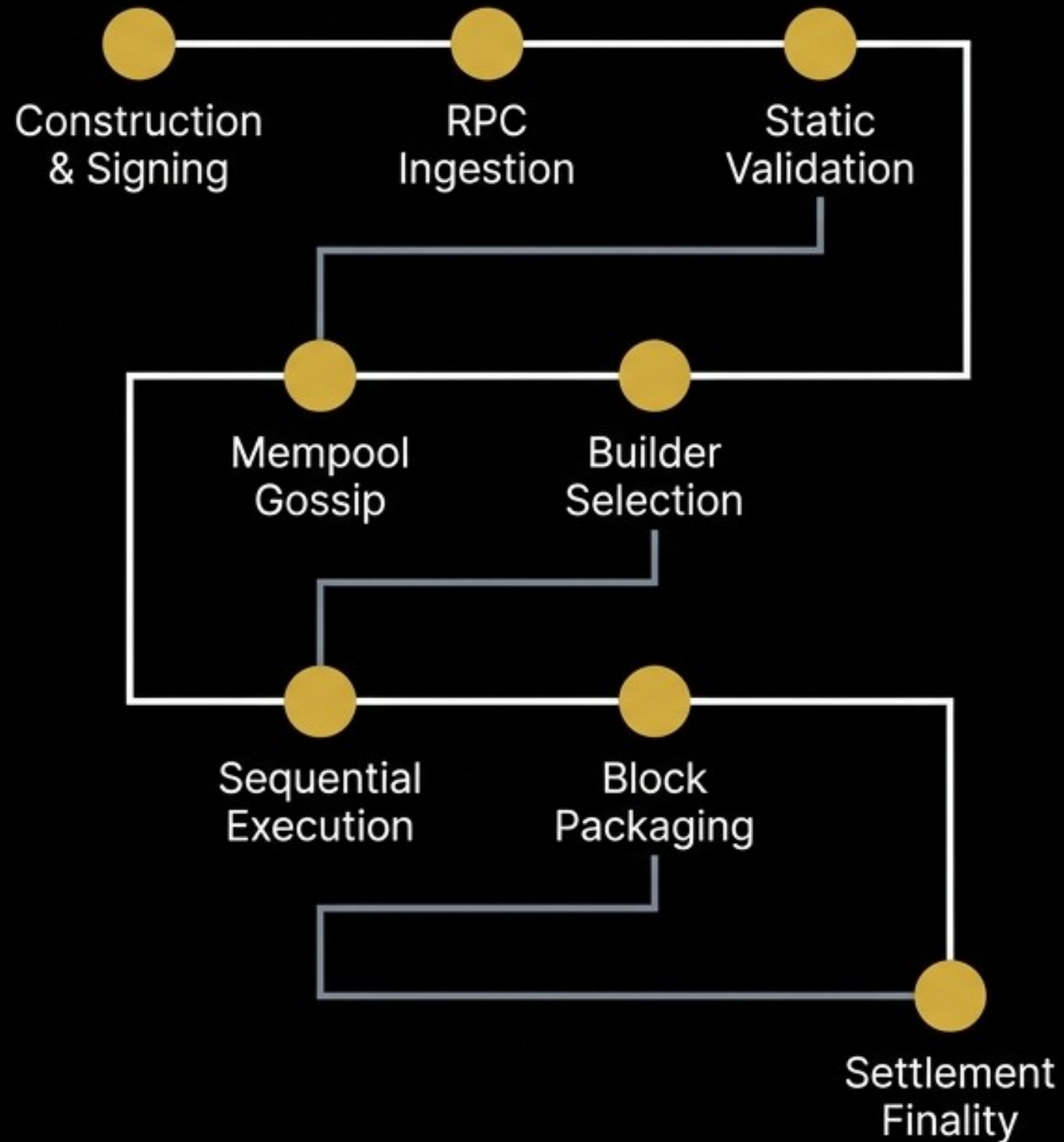




Transaction Lifecycle & State Transitions

A transaction is not a single event. It is a multi-stage distributed systems pipeline.

- **Local Realm:** Binary serialization and cryptographic signing at the device level.
- **P2P Network Realm:** RPC static validation and localized mempool queuing.
- **Consensus Realm:** Deterministic state mutation via the virtual machine.

Every latency spike, stuck transaction, and arbitrage vulnerability occurs within this precise sequence.

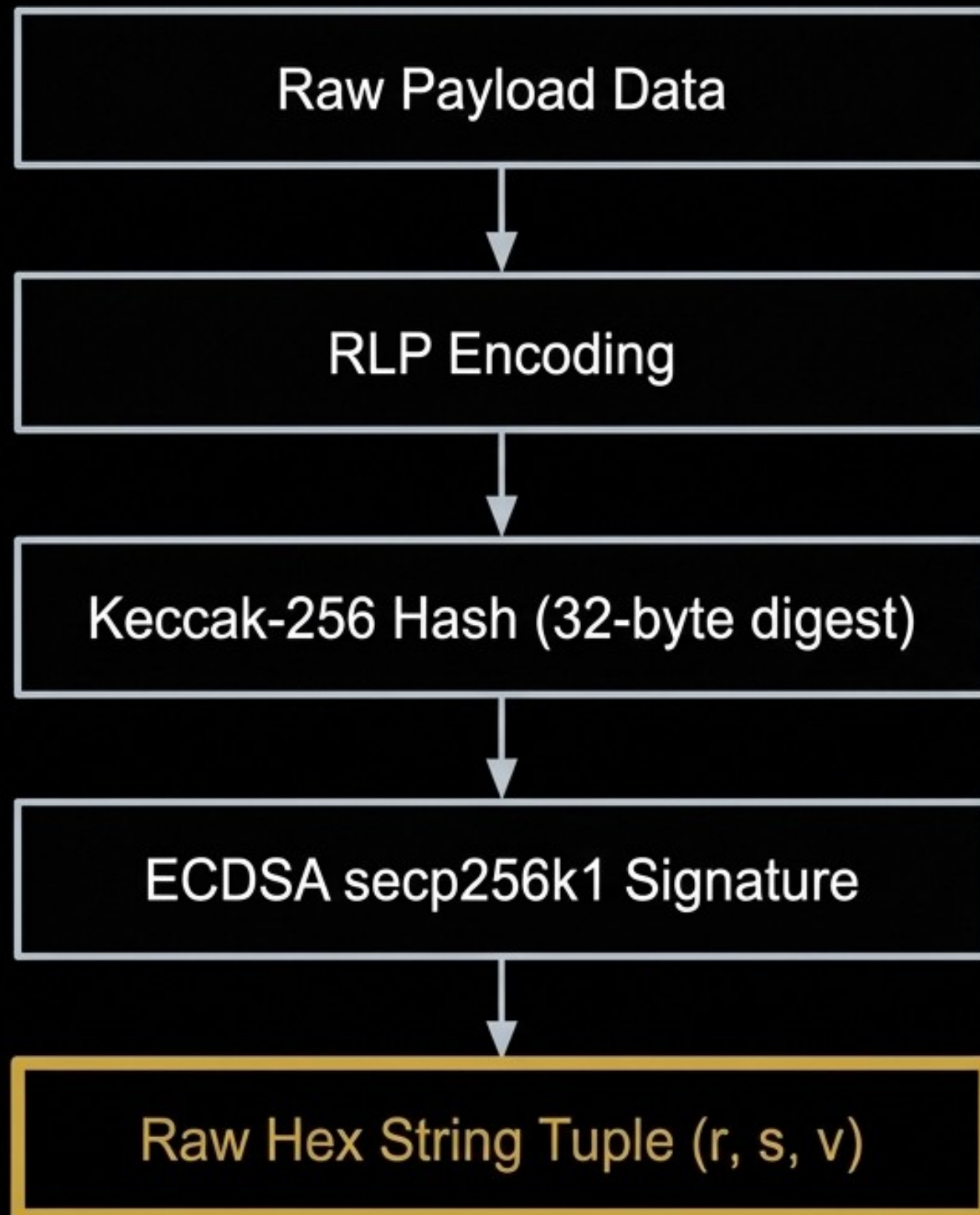
Phase 1: Payload Construction

Card 1

Core User Inputs		
to 20-byte destination address (EOA or smart contract).	value Native cryptocurrency transfer amount measured in atomic units.	calldata Arbitrary byte array. Empty for standard transfers; contains encrypted function selectors for contract execution.

Card 2

Auto-Injected Protocol Parameters (EIP-1559)		
gasLimit Maximum computational units authorized.	maxFeePerGas Absolute maximum price willing to be paid per gas unit.	maxPriorityFeePerGas Direct validator tip to incentivize rapid mempool selection.



Cryptographic Serialization & The Nonce

Before network broadcast, the wallet must assign a strict scalar integer: **the Nonce ($N_{tx} = N_{state}$)**.

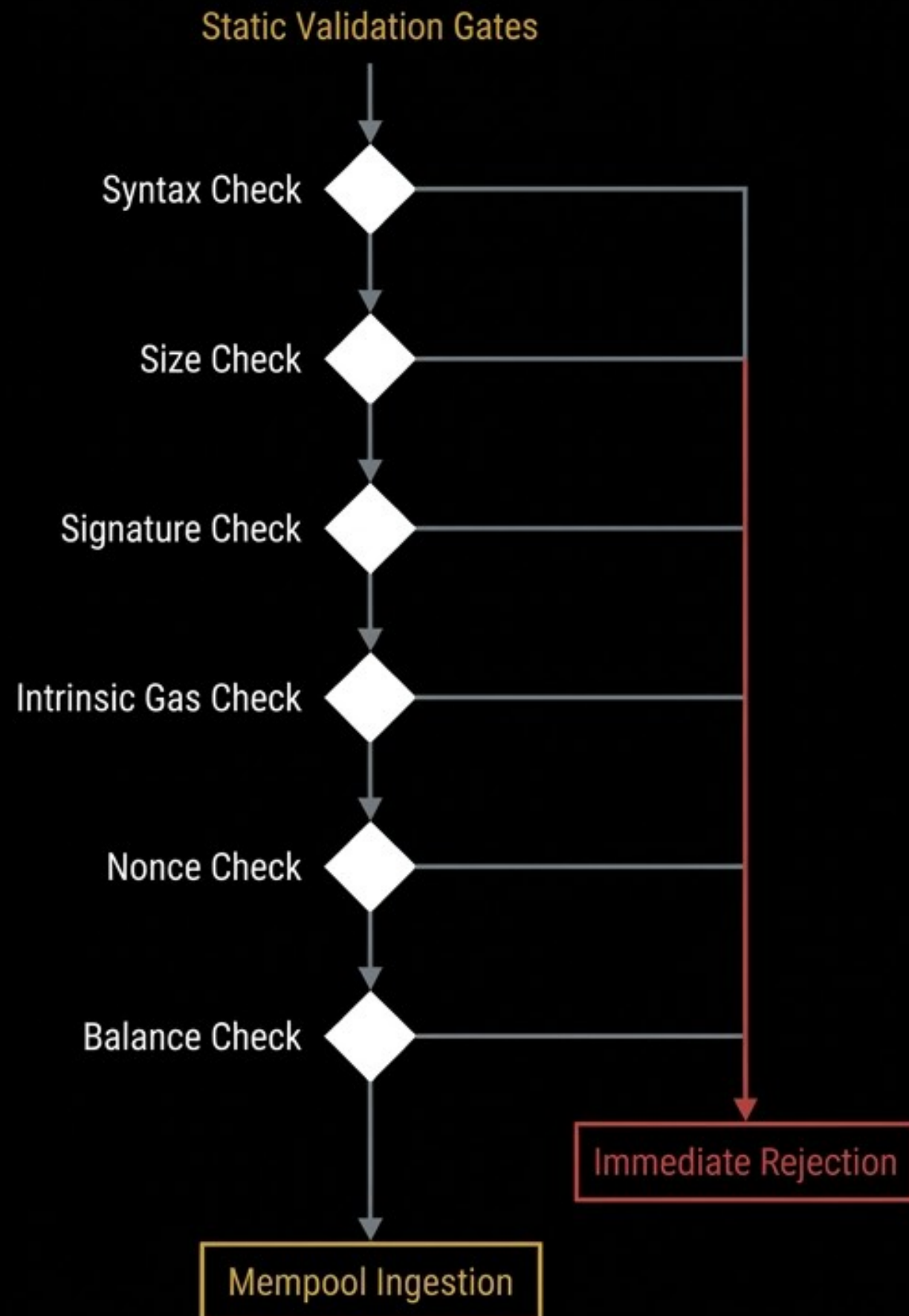
Unlike Bitcoin's mining nonce, an **Account Nonce** enforces two vital security guarantees:

1. Replay Protection

Rejects duplicated broadcasts of identical transactions.

2. Deterministic Sequencing

A transaction with nonce 5 is mathematically guaranteed to execute before nonce 6, regardless of validator arrival time.



Phase 2: RPC Ingestion

Payloads enter the network via the **eth_sendRawTransaction** JSON-RPC call.

RPC nodes do not blindly broadcast data. They act as the first line of defense, executing a 6-stage static verification sequence.

Transactions failing these checks are dropped in microseconds—expending zero miner gas and effectively neutralizing network-level DoS vectors.

$$G_{\text{intrinsic}} = 21000 + G_{\text{calldata}} + G_{\text{access_list}} + G_{\text{creation}}$$

Conditional: Contract Creation (+32,000 gas)

Variable Cost: Calldata
(4 gas per 0x00 byte; 16 gas per non-zero byte)

Base Cost: 21,000 Gas

Anatomy of Intrinsic Gas

Intrinsic Gas is the non-refundable computational baseline paid before the first opcode executes.

The 21,000 Base

Covers systemic node overhead—disk I/O operations, ECDSA signature recovery, and global account balance mutations.

Calldata Costing

Non-zero bytes cost 4x more than zero bytes (16 vs 4) specifically to penalize excessive permanent disk state bloat.

Local Node RAM

$N_{tx} = N_{state}$
Pending Queue

Nonce
10

Nonce
11

Ready for
Block Builder

$N_{tx} > N_{state}$
Queued (Future) Queue

Nonce
13

**Missing
Nonce 12
Gap**

Phase 3: Mempool Dynamics & Nonce Gaps

The mempool is not a synchronized global database; it is localized RAM queuing within individual nodes.

- **Pending Queue:** Transactions perfectly matching the current state nonce. Ready for immediate execution.
- **Queued Queue:** Transactions arriving out of sequence. A single missing nonce creates a nonce gap, paralyzing all subsequent transactions from that account indefinitely until the missing integer is mined.

1. Bot Front-run
(High tip, buys asset)

2. Victim Transaction
(Executes at severe slippage)

3. Bot Back-run
(Sells asset for risk-free profit)

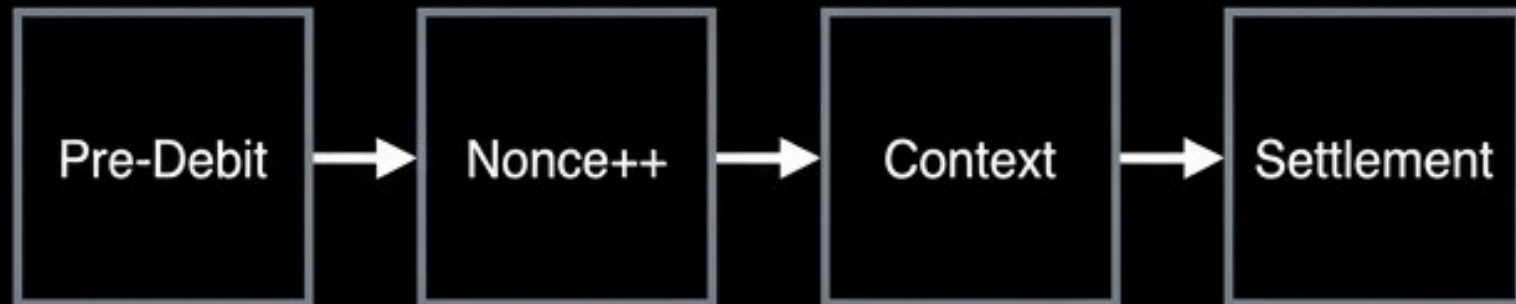
The Dark Forest & Replace-by-Fee

Public mempools are unencrypted. Arbitrage bots continuously scan this **Dark Forest** for extractable value (MEV), rearranging transaction execution orders via **validator tips** to execute sandwich attacks.

Resolving Gridlock: Users can unstick pending transactions via **Replace-by-Fee (RBF)**. By broadcasting a new payload with an identical nonce and a minimum **10% fee increase**, nodes will overwrite the paralyzed transaction in RAM.

$$\sigma_{t+1} = \Pi(\sigma_t, B_{t+1})$$

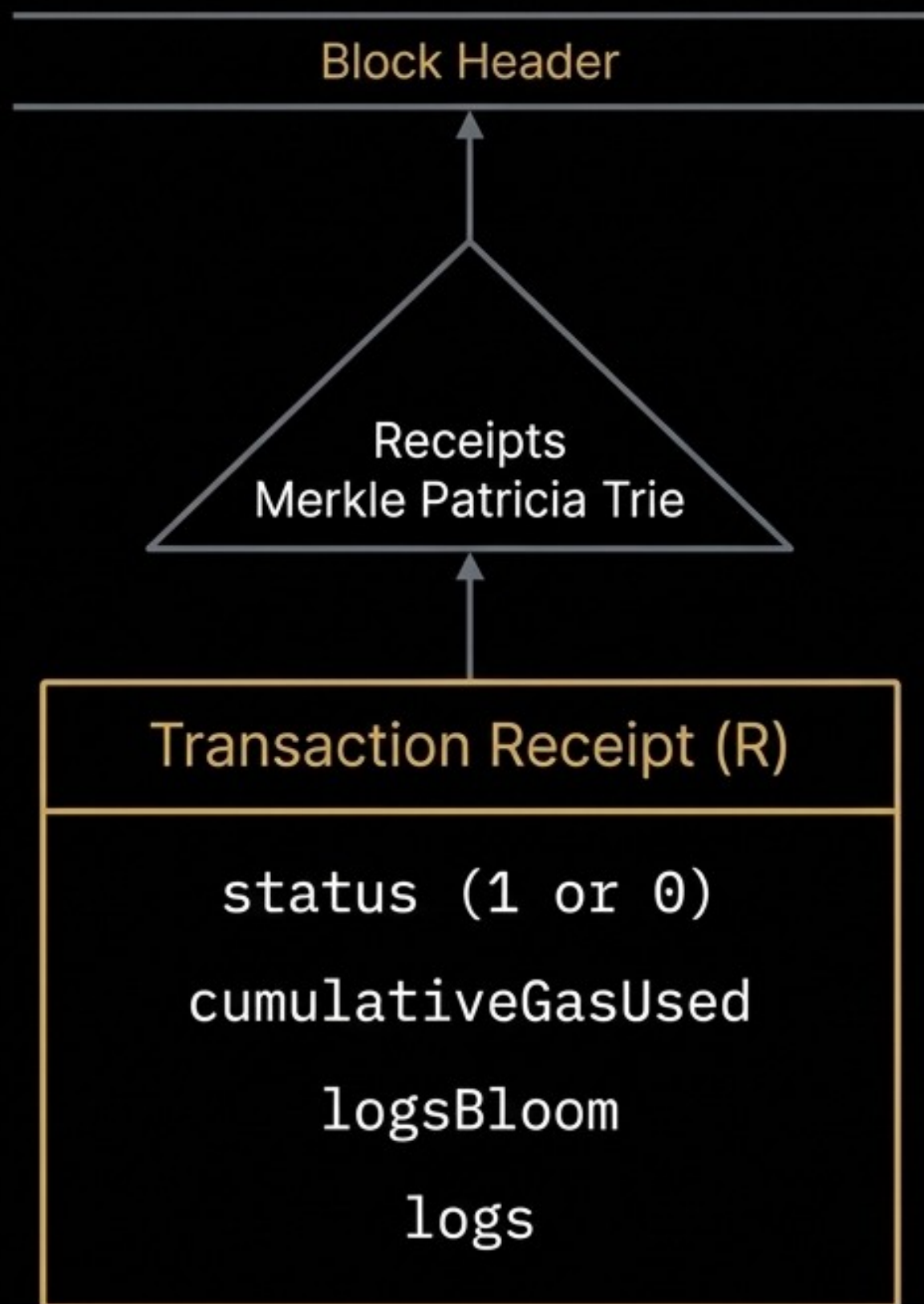
$$\sigma_i = \Upsilon(\sigma_{i-1}, T)$$



Phase 4: EVM State Transition

Block execution is a purely deterministic state machine. Each transaction processes via 4 atomic steps:

1. **Pre-execution Debit:** Sender balance is deducted ($\text{gasLimit} \times \text{EffectiveGasPrice}$) to guarantee solvency.
2. **Nonce Increment:** N_{sender} is incremented by exactly 1.
3. **Execution Context:** The EVM allocates stack memory and executes contract bytecode sequentially.
4. **Settlement & Refund:** Actual gas used is tallied. Unused gas is refunded, base fee is burned, and tip is credited to the validator.



Phase 5: Receipts & Event Logging

Upon execution, nodes generate an immutable proof: the Transaction Receipt.

Crucially, even if a transaction reverts mid-execution (status: 0), it is **permanently included** in the block and **gas is consumed**.

Because receipts are immutable, they are **NOT stored in the World State Trie**. They are aggregated into a dedicated **Receipts Trie**. Web3 frontends index the **logsBloom** within these receipts to update user interfaces without querying heavy global state.

Architecture Bridge: Ledger State Models

We have mapped the σ transition. But how is σ physically stored on disk?

The Physical Coin Model (UTXO)



Bitcoin Logic. The ledger functions as a pile of unspent digital cash bills. Transactions consume distinct inputs to forge entirely new outputs.

The Bank Ledger Model (Account State)

	—	323
	—	1,000
	—	500
	—	0

Ethereum Logic. The ledger functions as a centralized bank statement. Transactions mutate a global database of addresses and singular balances.

Next Module: Ledger State Models (UTXO vs. Account).